

TURMERIC PLANT

PROJECT REPORT

Internship at Olam Food Ingredients (ofi)



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ACKNOWLEDGEMENT

I extend my heartfelt gratitude to the following individuals whose support and guidance have been invaluable throughout this project:

Mr Jay Hari, Deputy General Manager at OFI, for his overarching guidance and for entrusting me with this opportunity.

Mr Saju, Assistant Manager, for his thorough explanations of the production process and invaluable assistance in understanding machinery and equipment.

Mr. Justin, Senior Executive, for his patient explanations and insights into electrical systems and wiring diagrams.

Mr. Bhema Teja, Senior Executive, for his excellent explanations about workplace safety. His expertise and guidance have been invaluable to this project

Mr. Abliash and **Mr. Rajesh**, for their expertise and invaluable mentorship.

These individuals have played crucial roles in the successful completion of this project, and I am sincerely thankful for their contributions.

Eldhose Babu
Intern -Project

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CHAPTER 1

INTRODUCTION

This report outlines my experience and contributions as a Project Engineer intern at Olam Food Ingredients (). The primary focus of my internship was the installation and commissioning of a turmeric processing plant.

1.1 GENERAL BACKGROUND

Olam Food Ingredients (ofi) is a new operating group born out of Olam, a leading supplier in the global food ingredients market. OFI offers sustainable, natural, value-added food products and ingredients, enabling consumers to enjoy the healthy and indulgent products they love. The group comprises industry-leading businesses in cocoa, coffee, dairy, nuts, and spices, and has built a unique global value chain presence, including its own farming operations, farm-gate origination, and manufacturing facilities. By partnering with customers and leveraging its complementary and differentiated portfolio of 'on-trend' food products, OFI co-creates solutions that anticipate and meet changing consumer preferences, particularly as demand increases for healthier and traceable food options.

During my internship at OFI, I had the privilege of being involved in a significant project aimed at establishing a new turmeric processing plant. This experience provided me with invaluable insights into the food ingredients industry and allowed me to develop a broad range of skills. Working on this project exposed me to the complexities of setting up a state-of-the-art processing facility, including the planning, design, and implementation phases. I

collaborated with a team of experienced professionals from various disciplines, including engineering, quality assurance, and supply chain management, ensuring that all aspects of the plant's establishment were thoroughly considered and executed.

1.2 SCOPE OF THIS PROJECT

The new project, Turmeric Plant 2, is set to revolutionize our production capabilities, doubling the current efficiency to an impressive 100 percent. This state-of-the-art facility incorporates optimized production times, ensuring faster turnaround without compromising quality. The integration of cleaner systems and full automation of all activities marks a significant advancement in our operations, promoting both environmental sustainability and operational excellence. Turmeric Plant 2 exemplifies our commitment to innovation and efficiency, positioning us at the forefront of the industry.

1.3 NEED OF THIS PROJECT

The existing grinding system is only 150 Kg/hr and will not handle bulbs. The new system will be able to grind bulbs at the capacity of 500 Kg/hr with 250/300 microns. The new layout will be more spacious and with separate feeding and FG collection points.

1.4 OBJECTIVE OF THIS INTERNSHIP

- Gain practical experience in project engineering.
- Understand the process flow of a turmeric processing plant.
- Learn about the installation and commissioning of industrial

CHAPTER 2

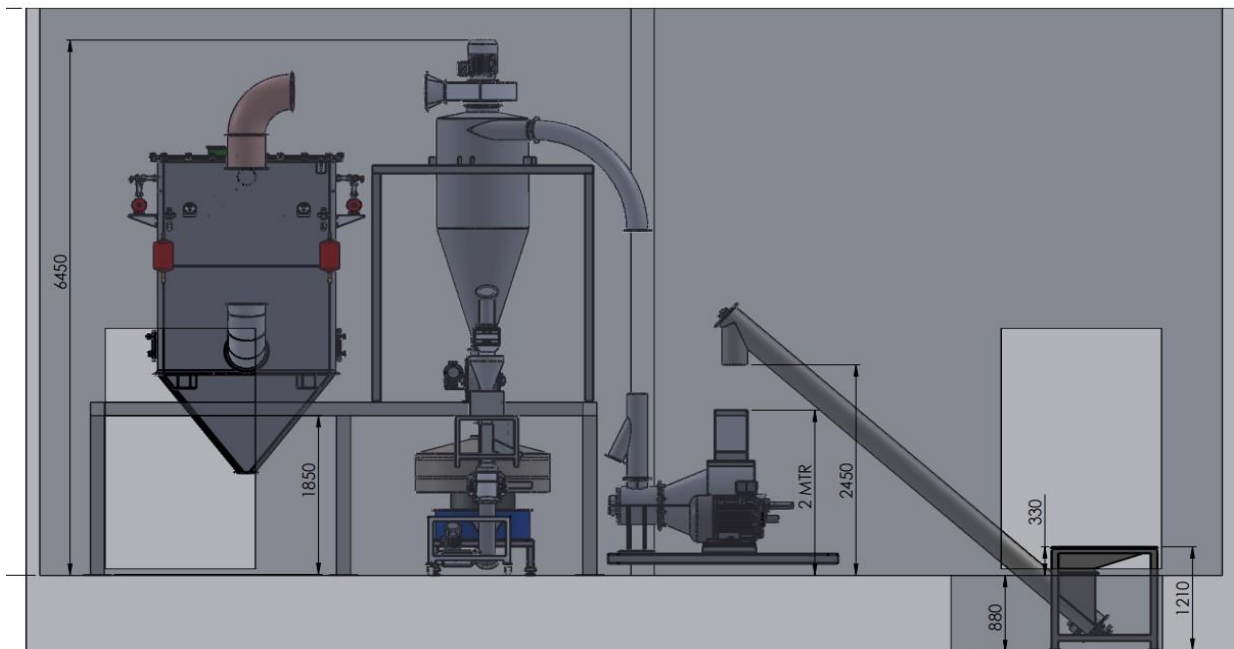
METHODOLOGY

The methodology followed for the project involved the following steps:

PLANNING AND PROPOSAL:

- **Purpose:** To define the project scope, objectives, and deliverables.
- **Activities:** Initial brainstorming, feasibility assessment, defining timelines and budgets, and preparing a formal proposal for approval.

DESIGN:



Purpose: Create CAD drawings for plant layout.

Activities: Engineering design, layout planning, CAD drafting.

Outcome: Detailed plant layout blueprint for construction.

Fig 2.1

Basic CAD drawing of the plant

PROCUREMENT:

- **Purpose:** Acquire machinery and equipment.
- **Activities:** Identify needs, source suppliers, obtain quotes, negotiate contracts, procure items.
- **Outcome:** All necessary equipment and materials acquired.

INSTALLATION:

- **Purpose:** Erect machinery and set foundations.
- **Activities:** Machinery assembly, foundation laying, electrical installation.
- **Outcome:** Completed plant installation ready for operation.

COMMISSIONING:

- **Purpose:** Ensure plant functionality.
- **Activities:** Conduct trial runs, test equipment, adjust settings.
- **Outcome:** Fully operational plant ready for production.

CHAPTER 3

PROJECT OVERVIEW

The turmeric plant was designed to process raw turmeric through a series of mechanical and filtration steps to produce a high-quality finished product. The main components and their functions are described below:

3.1 SCREW FEEDER

We installed the screw feeder for feeding raw turmeric into the production line by first excavating a pit with a depth of 800mm. The feeder, which stood 1210mm high with a base measuring 330mm by 330mm, was carefully positioned and aligned with the production line. We secured the feeder in place within the pit to ensure stability and connected the outlet to the production line to prevent turmeric spillage. Electrical connections were made according to safety standards, and all mechanical components were securely fastened. After installation, we conducted thorough testing and calibration to ensure the feeder operated correctly and efficiently. Maintenance access and vibration isolation were also considered to ensure long-term reliability and ease of service.

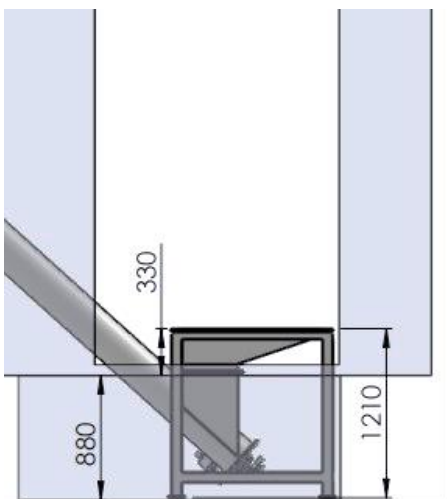




FIG INSIDE VIEW OF SCREW FEEDER

3.2 IMPACT MILL

Impact Mill is the heart of our production line. Driven by a robust 100 hp motor, this advanced equipment efficiently grinds material fed through its hopper and discharges it into a cyclone for optimal separation. With dimensions of 3 meters by 2 meters, the impact mill is securely mounted on vibrating pads to minimize noise and vibration, ensuring a smooth and efficient operation

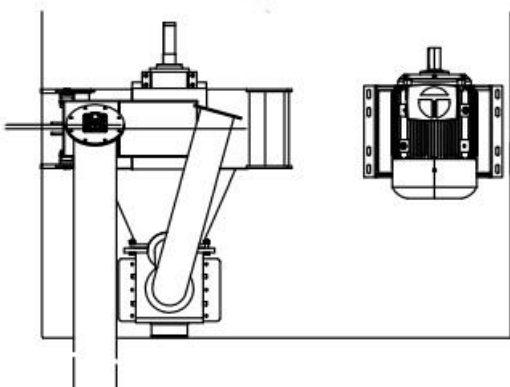


Fig: Inside view of mill – During trail ru

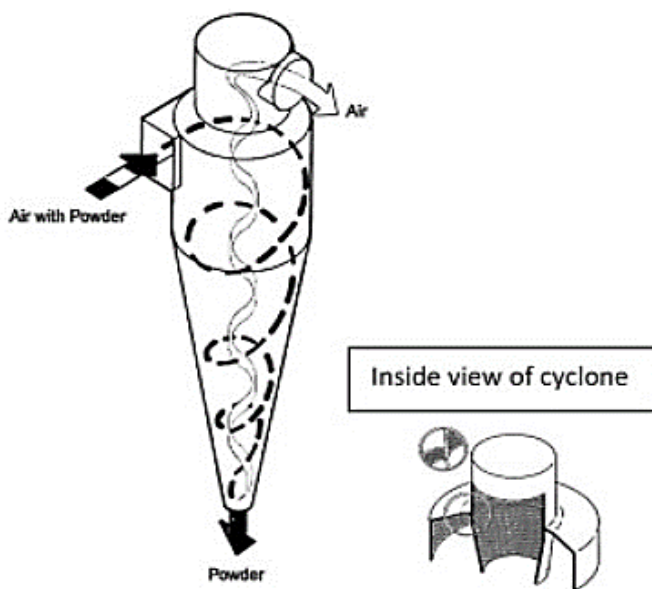
3.3 CYCLONE

A cyclone separator is a vital component in industrial processes for efficiently separating solid particles from light particles. Operating on the principle of centrifugal force, the cyclone forces the powder mixture into a spiral motion upon entry, creating a vortex inside the cylindrical barrel. This vortex causes heavier solid particles to move outward and downward due to centrifugal force, collecting at the conical bottom of the separator. Meanwhile, the lighter powder particles, along with finer particles, moves towards the center and exits through an outlet at the top. This design ensures effective separation, with the cyclone's construction facilitating gravitational settling of solids in the conical section for easy removal.



Commonly integrated with a powerful blower to induce the necessary airflow, cyclone separators are employed across various industries such as pharmaceuticals, mining, and food processing. They play a critical role in dust collection systems, improving air quality by efficiently capturing and separating particulate matter from industrial exhaust streams. The simplicity and effectiveness of cyclone separators make them indispensable in maintaining clean and safe working environments while enhancing overall process efficiency in diverse manufacturing applications.

Fig 3.3



3.4 AIR LOCK VALVES

An airlock rotary valve is essential in the food industry for managing the flow of bulk materials between different processing stages while minimizing pressure loss. The valve features several critical components that work together to ensure efficient operation.



COMPOENTS

- **Housing:** Made of stainless steel, it confines the bulk material, ensuring safe handling throughout the process.
- **Rotors:** These metering components consist of a shaft and blades (vanes) welded or cast together, forming rotor pockets that transport the material. The number of vanes can vary (e.g., 6 or 8).

- **End-Plates:** They prevent bulk materials from spilling over the sides of the rotor pockets.
- **Systems of Propulsion:** This includes a 1.5 hp electronically induced motor and a reduction gearbox that together drive the rotor's rotation.

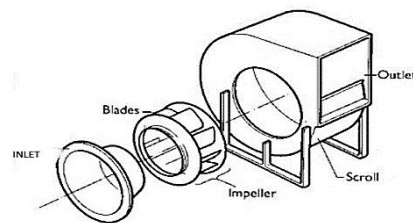


INSIDE VIEW OF VALVES

3.5 CENTRIFUGAL BLOWER'S

A radial centrifugal blower is a mechanical device used to move powder into a cyclone by means of vacuum energy. The blower features a rotating impeller with radially arranged blades that create the vacuum needed to transfer the powder from the impact mill into the centre through the inlet of the cyclone. As the impeller spins, vacuum is induced in the cyclone, efficiently increasing flow and vacuum pressure, making radial centrifugal blowers highly effective for this application.

In this plant, a 15 hp blower is used for the cyclone, ensuring sufficient vacuum energy to move the powder efficiently. Additionally, a 3 hp blower is utilized for the filter bag, providing optimal performance for the dust collection stage of the process. This dual-blower setup ensures that each stage of the powder handling process operates at peak efficiency, contributing to the overall



effectiveness of the plant's operations.



Airflow (CFM) = Fan Speed (RPM) x Fan Blade Diameter (in.) x Fan Pitch (in.) x $\pi/6$.

3.6 ROTARY SIFTER

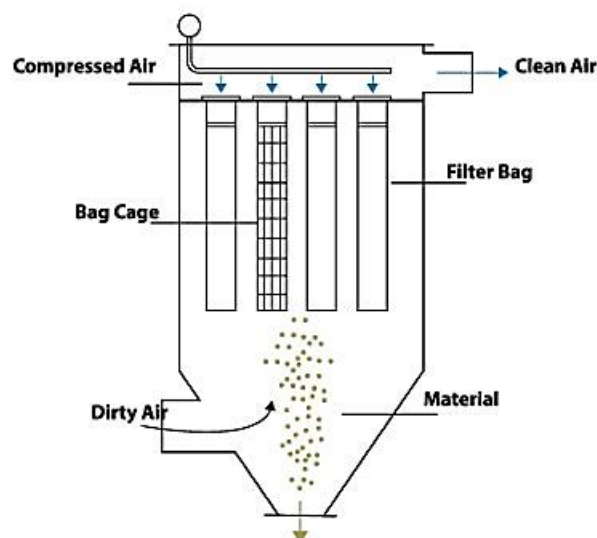
The rotary sifter is used to separate and grade food products based on their size and shape. The working principle involves the use of centrifugal force to achieve particle separation. Material is fed into the sifter from the top, where it is evenly distributed on the cylindrical screen. As the screen rotates, the material experiences centrifugal force, causing smaller particles to pass through the screen's mesh openings while larger particles are retained on the screen, ensuring efficient and precise sorting.



3.7 PULSE JET COLLECTOR

A pulse-jet dust collector is a system used to filter dust particles from an air stream, ensuring clean air is released into the atmosphere. It consists of one or more compartments, each containing rows of cylindrical fabric bags or pleated cartridges that act as filters. As dust-laden air enters the dust collector, it flows along the surface of the bags and then passes radially through the filter media. The dust particles are trapped on the outside of the bags, while the cleaned air passes through and is vented out.

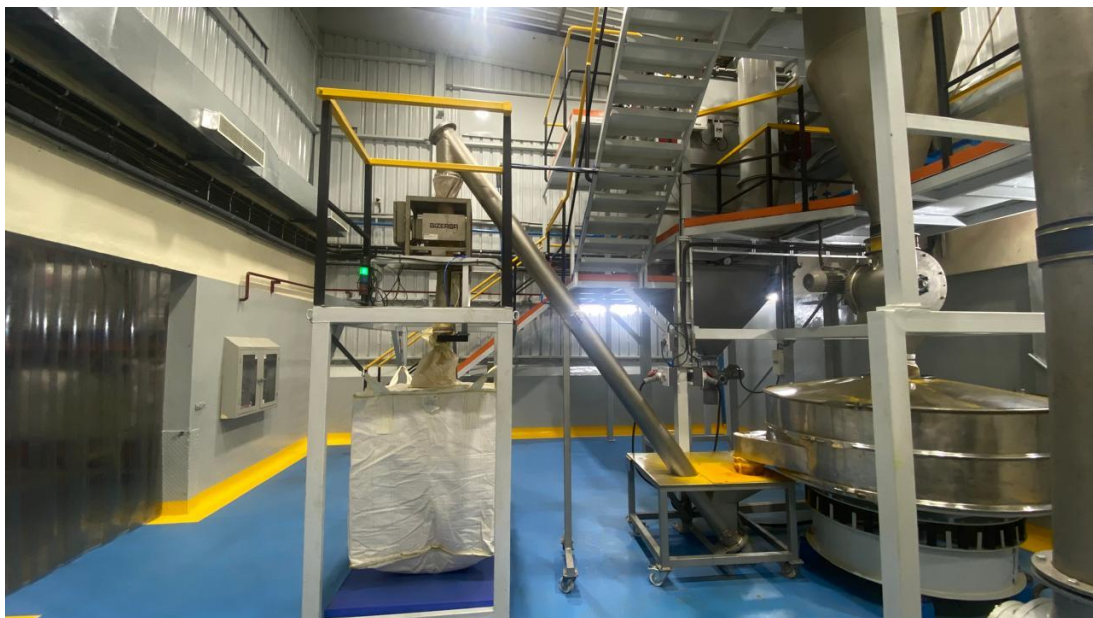
When the dust accumulation on the bags reaches a certain level, causing a pressure drop, the system activates a cleaning mechanism. This mechanism sends a pulse of compressed air through each row of bags, shaking off the collected dust. The dislodged dust falls into a hopper at the bottom of the collector and is then discharged through a Rotary Airlock Valve (RAL), ensuring a continuous operation without leakage. The collected layer of dust, known as the dust cake, enhances the filtration efficiency, allowing the dust collector to achieve a high efficiency rate of 99.9% for capturing very fine particles.



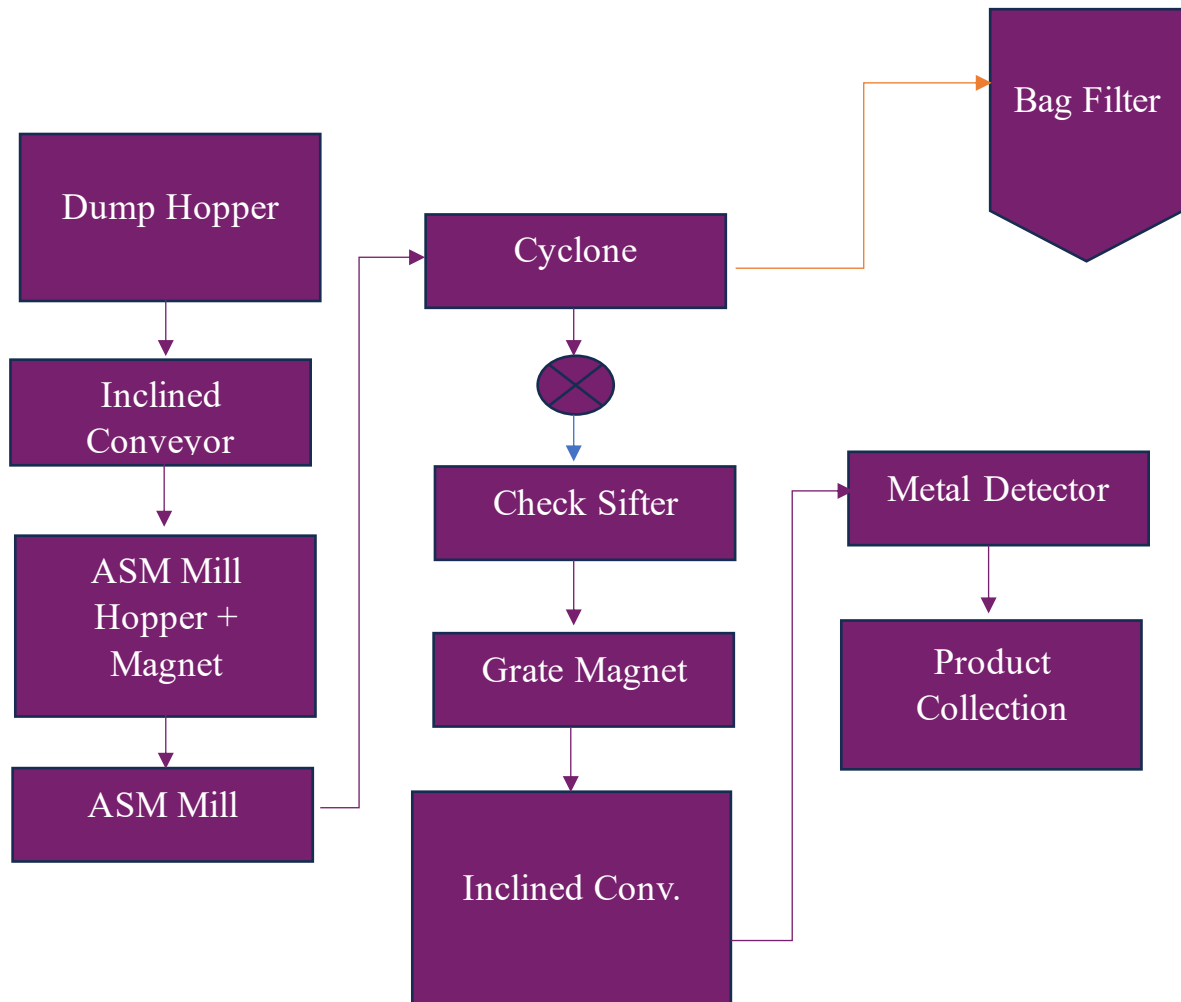
3.8 COLLECTION POINT

At the packing point, materials are first conveyed to a metal detector, which checks for any metallic contamination. If contaminants are detected, the deviator sorts out the contaminated materials, ensuring they do not proceed further. Clean materials continue on the conveyor to the bagging station, where they are filled into jumbo bags. Each bag is then sealed and labelled with relevant information such as batch number, weight, and material type. The sealed bags are transported to the storage area, ready for dispatch. Throughout the process, regular calibration of the metal detector, routine maintenance of the deviator and conveyor system, and strict quality assurance protocols ensure that only high-quality, contamination-free materials are packaged and dispatched. Detailed logs and inspection records are maintained to ensure compliance with safety and quality standards.

FIG : COLLECTION POINT



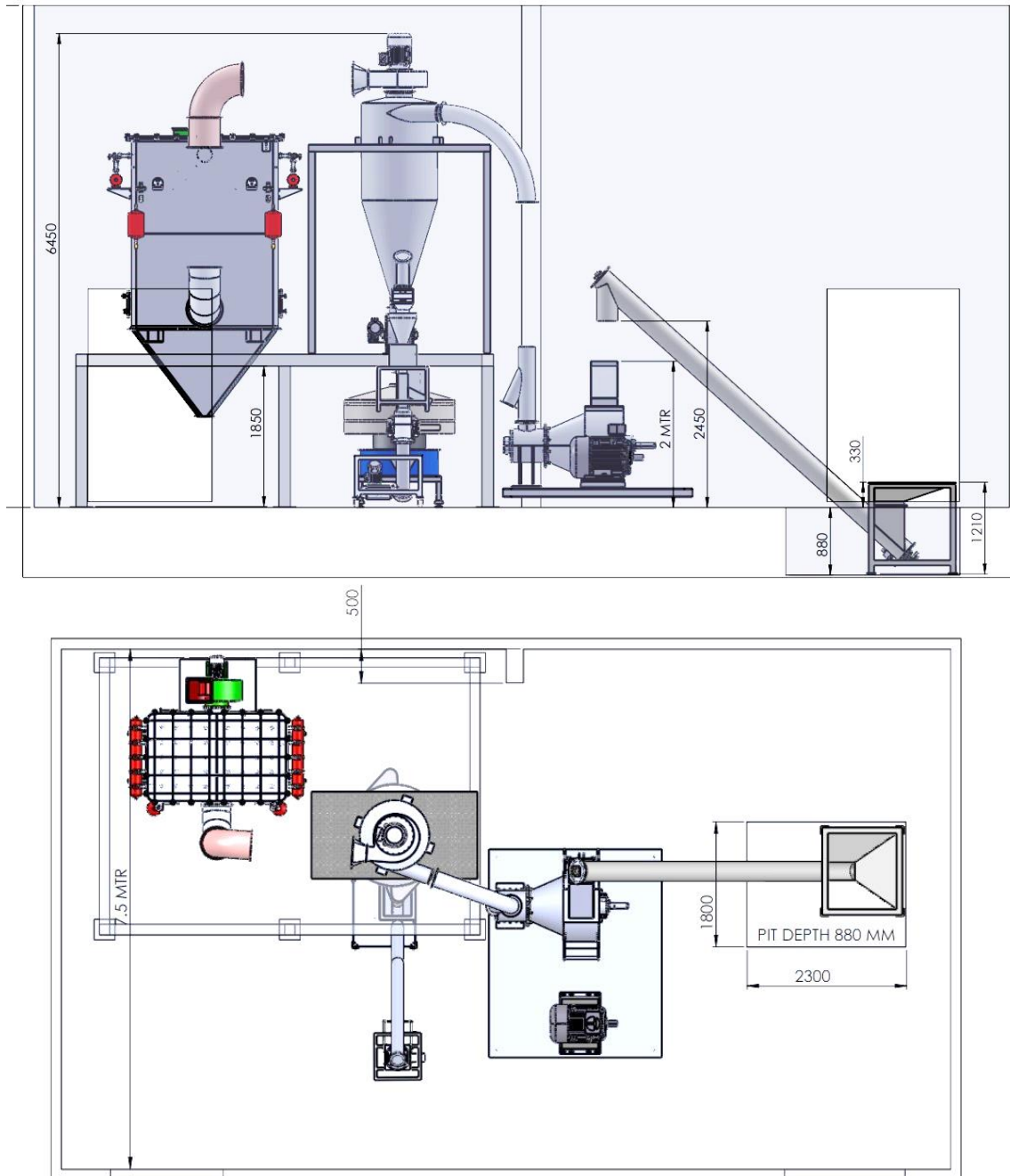
PROCESS FLOW DIAGRAM

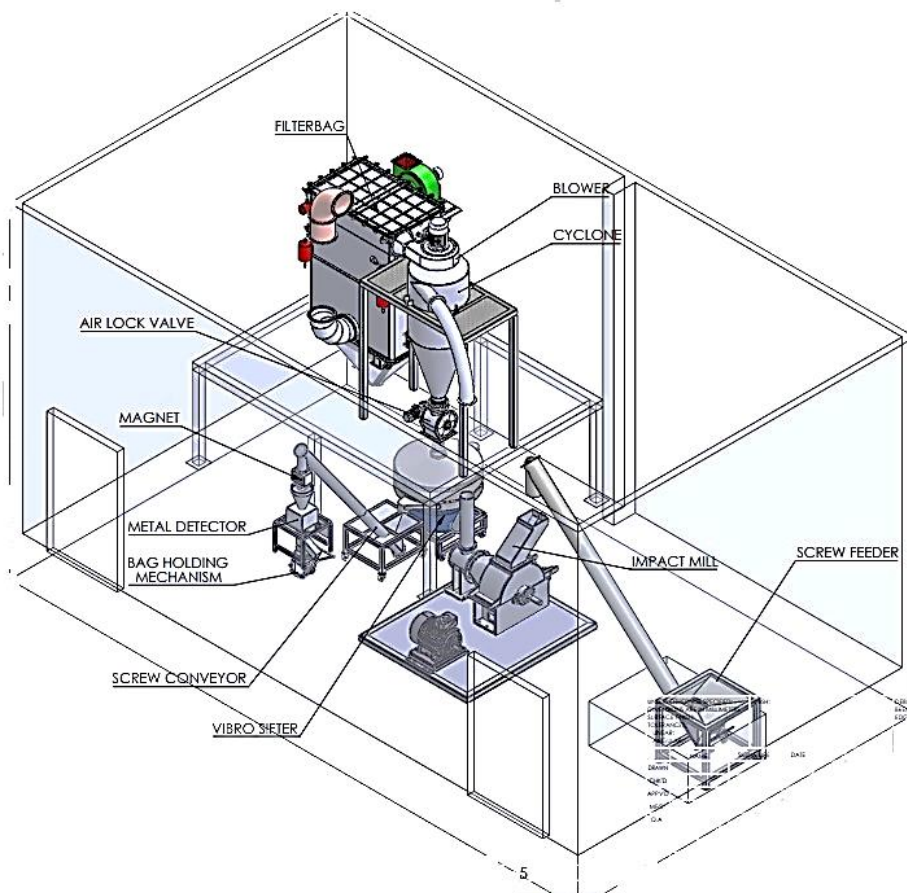
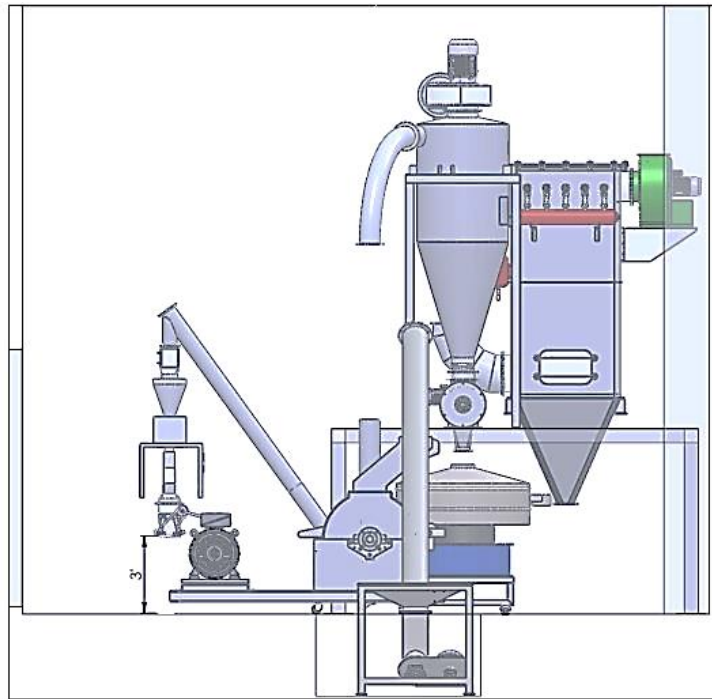


CHAPTER 4

PROJECT SCHEDULE

4.1 PLANNING AND PREPARATION





From (May 1st - May 7th)

Project planning and proposal were completed.
CAD drawings were designed.

4.2 MACHINERY PROCUREMENT

From (May 8th - May 13th)

Gathered all machinery equipment's for the turmeric plant



Fig -Glimpse of materials arrived

Gathered and Stored all machinery equipment's for the plants such as

FEEDER

CYCLONE

ROTARY VALVES

SCREW CONVEYORS

FILTER BAGS

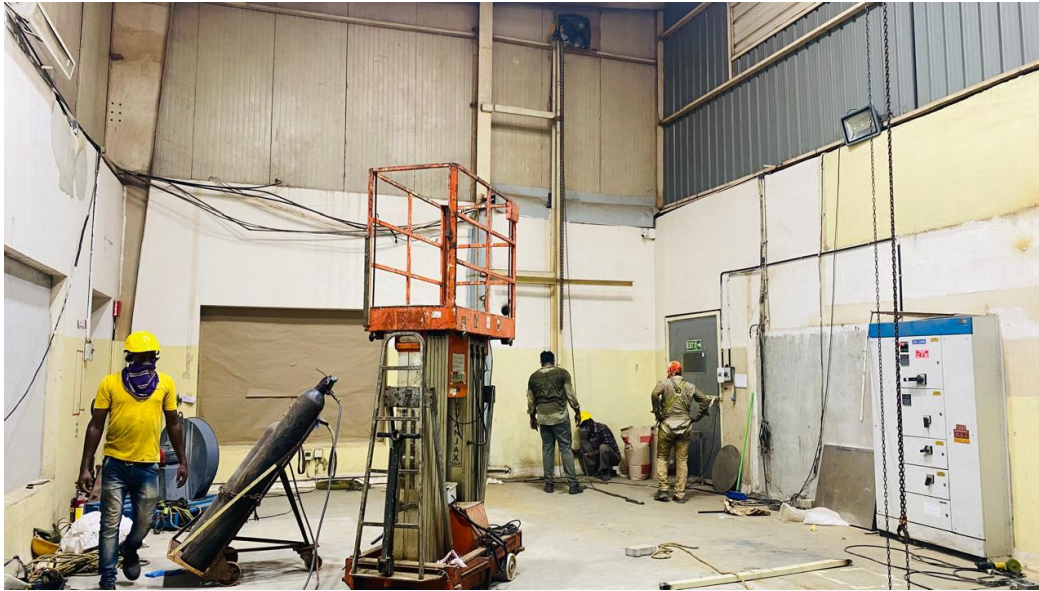
VFD'S

4.3 LAYOUT AND ERECTION

From May 14th to May 23rd

CLEARING THE OLD AREA:

- Cleared the designated area to make space for the new plant setup.
- Removed old equipment, debris, and any obstacles that could interfere with the installation of new machinery.



OUTLINING MACHINERY PLACEMENT:

- Carefully planned and marked the locations for each piece of machinery.
- Ensured optimal placement for efficient workflow and accessibility.

PIT DIGGING FOR FEEDER PLACEMENT:



- Excavated pits for the placement of feeders.
- Ensured proper dimensions and alignment to accommodate the feeders securely and facilitate seamless material flow into the processing line.

4.3.1 FLOOR GRINDING AND FLOOR CUTTING



4.3.2 MACHINERY ERECTION





Fig : Planning of daily activities



Fig: installing of air purging

4 ELECTRICAL WORKS AND ADDITIONAL PLAT FROM WORKS

Finished on the dates - (May 24th - June 2nd)

- Conducted electrical installations.
- Constructed handrails, chequer plates, and set machinery foundations.
- Installed Pannel board.

LINKING CABLES TO PANEL BOARD



PANEL BROAD



CABLE TARY AND CABLES

VFD LINKING

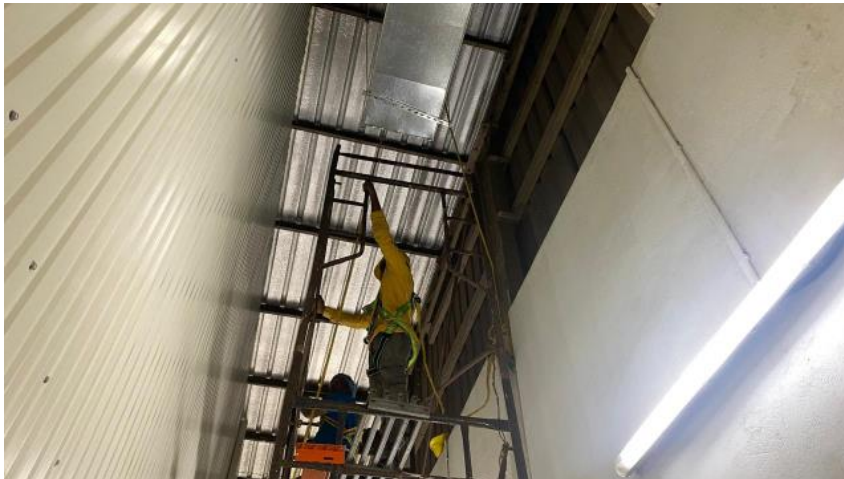


4.4.2 ADDITIONAL PLATFORMS AND HAND RAILS



4.5 VENTILATION WORKS (*JUNE 3RD - JUNE 7TH*)

- Installed air in ducts and air out ducts.
- Installed fans at the end of both ducts



AIR VENTILATION WORKS



4.6 PU COATING ,FLOORING AND PAINTING

Finished in the following dates (JUNE 7TH -10TH)

Applied PU coating on walls using FK111.
Completed flooring with R70CP material.



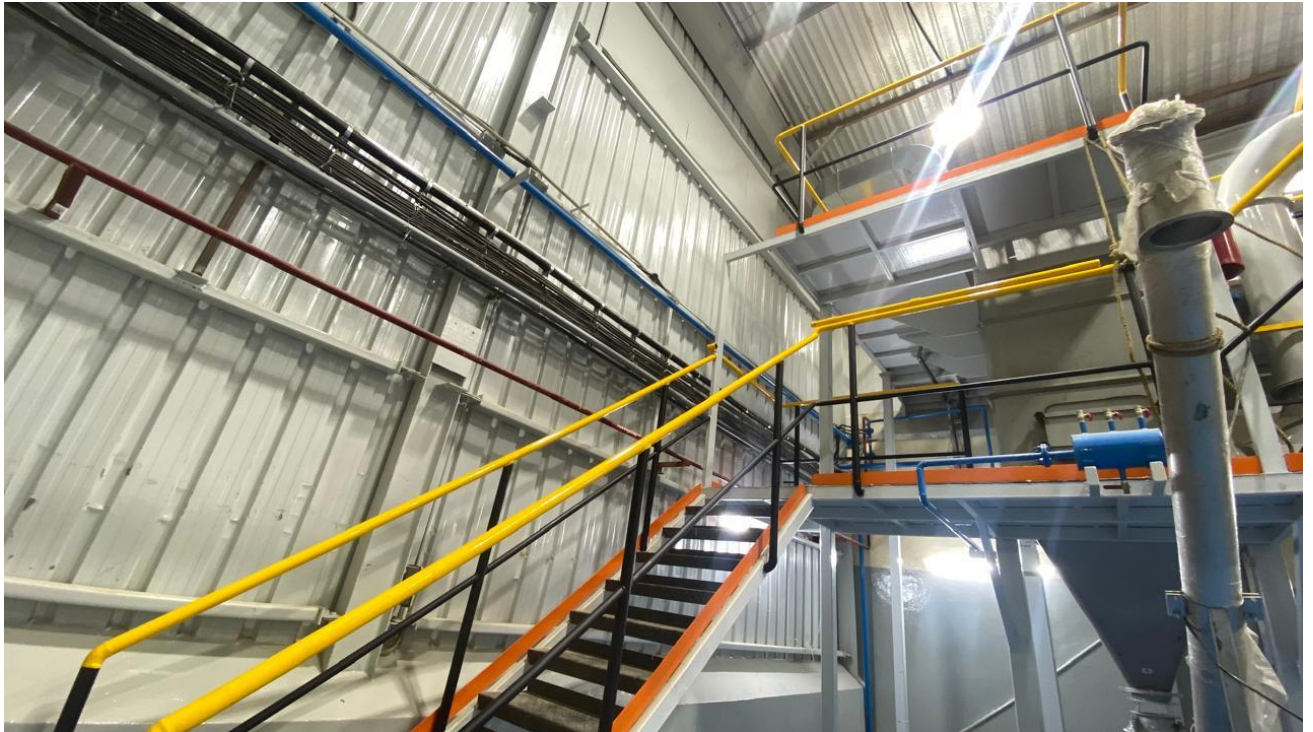
PU COATING ON WALL



PU FLOORING

4.6.1 PAINTING WORKS

- Painted safety rails: yellow on the top handrail and orange on the lower part.



HAND RAILS PAINTED

4.7 TRAIL RUN

Trail run was successfully completed from the dates - JUNE(13TH -19TH)

On trail run we got required spec – 300microns

For this we adjusted the impact mill (classifier) also the angle of swifter

Total output on trail days -1100 kg



TRAIL RUN – FEEDING MATERIAL INTO SCREW FEEDER



CHAPTER 5

LIST OF MATERIALS

TABLE 5.1 LIST OF MECHANICAL EQUIPMENTS

SI NO	ITEM	POWER (HP)	NO OF MACHINE	NO OF MOTORS	TOTAL POWER
1	8 inch Screw Conveyor (Reverse forward with VFD)	5 HP	1NOS	1	5
2	36 x 18 Grinding Mill (VFD)	100 HP	1NOS	1	100
3	Lifting Blower	15 HP	1NOS	1	15
4	Hydraulic Lifting	2 HP	1NOS	1	2
5	Rotary Valve	1.5 HP	1NOS	1	1.5
6	40 Nos Filter bag assembly (single Phase Neutral)	3 HP	1NOS	1	3
7	60 " Vibro- Sifter	2 HP	1NOS	1	2
8	Packing Conveyor	1.5 Hp	1NOS	1	1.5
9	150 Dia Metal Detector	230 V I phase Industrial Power plug	1NOS	1	230 V I phase Industrial Power Plug

TABLE 5.2 LIST OF ELECTRICAL QUANTITY

SI NO	ITEM NAME	QTY	UNIT
1	LT PANEL BOARDS	1	Nos
2	CABLE TRAY		
2.1	300mm Ladder	50*700	m
2.2	400mm Ladder	50*830	m
3	COPPER CABLES		
3.1	3.5C x 2.5 sq.mm	275*240	m
3.2	3.5C x 1.5 sq.mm	50*183	m
3.3	1C x 95 sq.mm	90*1240	m
4	GLANDING		
4.1	0.35C x 240 sq.mm	2*200	Nos
4.2	0.35C x180 sq.mm	2*1750	Nos
4.3	0.35C x10 sq.mm	4*850	Nos
4.4	0.35C x2.5 sq.mm	12*550	Nos
4.5	0.35C x 1.5sq.mm	16*275	Nos
4.6	95 sq.mm 6*275	6*275	Nos
5	MS FABRICATION		
5.1	500*140 MS	500*140	Nos
6	EARTH		
6.1	Cu SWG No 10	275*140	m
6.2	Cu Bench 25x2mm	8*500	m
6.3	Cu 25x2mm	100*910	m
7	EXISTING MSB MODIFICATION		
7.1	MSB Modification	1	Nos
8	RTCC PANEL SHIFTING		
9	ISOLATOR IN ENCLOSURE (Local isolation of motors)	6	Nos
10	TERMINATION		
10.1	3.5x240 sq.mm 100*1310 Al Ar cable	100*1310	m
10.2	3.5x10 sq.mm Al Ar cable	1	Nos
11	SAFETY EQUIPMENTS		
11.2	Supply and Laying of 1.1 KV Grade Elastomer Polymer 2mm thickness	2	m
11.3	Emergency trip push button	2	Nos

CHAPTER 6

WORK SAFETY

6.1 WORK SAFETY AND PERMITS

For our recent project, we have implemented a comprehensive array of work permits to uphold safety standards and regulatory compliance. The General Work Permit serves as the cornerstone, facilitating a wide range of tasks from routine maintenance to complex construction activities while strictly adhering to safety protocols. This permit ensures that all operations are conducted with the utmost caution and adherence to procedural guidelines, thereby minimizing risks and ensuring operational efficiency.

In addition to the General Work Permit, specialized permits have been issued to manage specific high-risk scenarios effectively. The Confined Space Work Permit is obligatory for tasks within enclosed or partially enclosed spaces, addressing hazards such as restricted access and confined movement. Our Height Work Permit is essential for activities performed at elevated locations, mitigating potential fall hazards through rigorous safety measures and equipment protocols. Moreover, the Hot Work Permit is crucial for operations involving activities like welding and cutting, ensuring stringent fire prevention measures are in place to safeguard personnel and infrastructure.

Furthermore, all project personnel undergo mandatory safety induction training covering fundamental workplace safety principles. This training encompasses the correct usage of personal protective equipment (PPE), hazard identification techniques, emergency response procedures, and best practices for safe working conditions. By equipping our workforce with these essential skills and knowledge, we cultivated a culture of safety consciousness across all project activities, significantly reducing the incidence of accidents and promoting a secure working environment.

CHAPTER 7

CONCLUSION

The completion of Turmeric Plant 2 marks a significant milestone for Olam Food Ingredients (OFI) and embodies our commitment to innovation, efficiency, and sustainability in food processing. Throughout this project, I have had the privilege of contributing as a Project Engineer intern, gaining invaluable practical experience in industrial engineering and project management.

The journey from initial planning and procurement to installation and commissioning has been both challenging and rewarding. We have successfully integrated state-of-the-art machinery including screw feeders, impact mills, cyclones, air lock valves, centrifugal blowers, rotary sifters, pulse jet collectors, and more, each playing a crucial role in optimizing our turmeric processing capabilities.

The project timeline, meticulously planned and executed, ensured that each phase—from site preparation to electrical and ventilation installations, PU coating, flooring, and final testing—was completed with precision and efficiency. This approach not only streamlined operations but also enhanced safety standards, as evidenced by the comprehensive safety protocols implemented throughout the project duration.

Moreover, the support and guidance from Mr. Jay Hari, Mr. Saju, Mr. Justin, Mr. Bhema, Mr. Abliash, and Mr. Rajesh have been instrumental in overcoming challenges and achieving our goals. Their expertise, dedication, and mentorship have significantly contributed to the successful implementation of Turmeric Plant 2.

Looking forward, this project sets a benchmark for OFI, showcasing our ability to innovate and adapt to industry demands while upholding the highest standards of quality and sustainability. As we continue to expand our capabilities and product offerings, Turmeric Plant 2 stands as a testament to our commitment to excellence and our vision for the future of food ingredient processing.

In conclusion, I am grateful for the opportunity to be part of this transformative project at OFI. It has not only deepened my technical skills but also instilled in me a profound appreciation for teamwork, diligence, and innovation in the pursuit of operational excellence. I look forward to applying these experiences and insights as I continue to grow in my career as a professional engineer